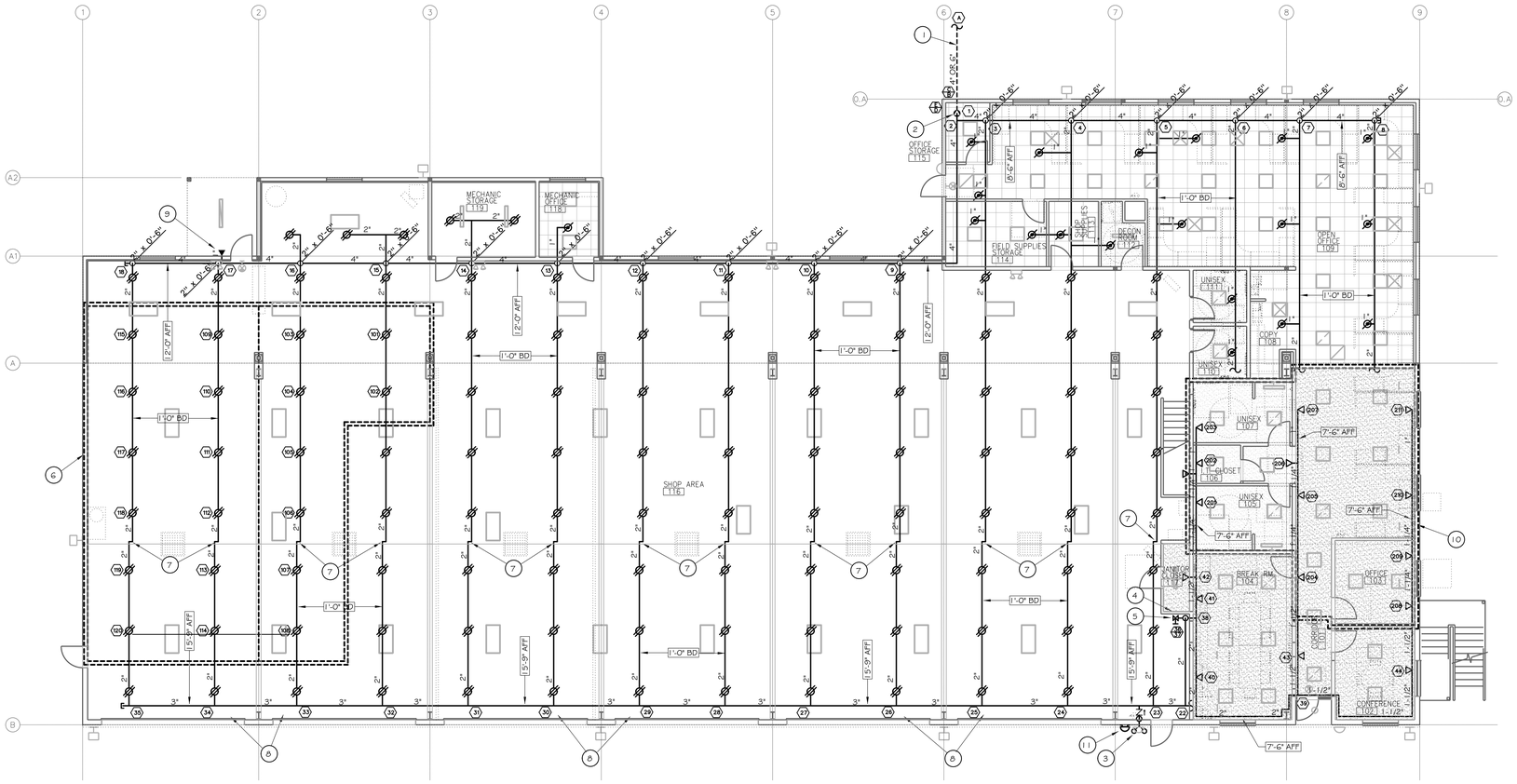
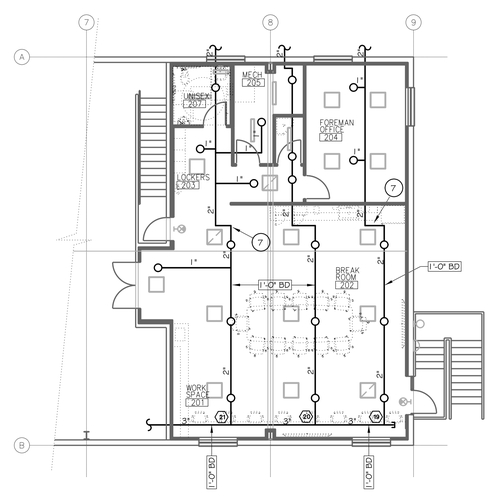


1 FIRE SPRINKLER PLAN – MAIN FLOOR
SCALE 1/8" = 1' – 0"



2 FIRE SPRINLER PLAN – MEZZANINE
SCALE 1/8" = 1' – 0"

FIRE SPRINKLER KEY NOTES

1. NEW 4" OR 6" UNDERGROUND FIRE SPRINKLER WATER SUPPLY TO BUILDING BY OTHERS. WATER SUPPLY TO BE PROVIDED BY 28,000 GALLON WATER STORAGE TANK AND 500 GPM ELECTRIC FIRE PUMP. WATER SUPPLY SHALL PROVIDE A MINIMUM OF 60 PSI AT 500 GPM FOR A DURATION OF 60 MINUTES.
2. WET-PIPE FIRE SPRINKLER RISER. SEE DETAIL 1 ON SHEET F5-2.
3. 2-WAY, WALL MOUNTED FIRE DEPARTMENT CONNECTION. MOUNT AT 36" ABOVE FINISHED EXTERIOR GRADE. CONNECT TO 3" FIRE SPRINKLER CROSS MAIN THAT RUNS ALONG SOUTH WALL OF BUILDING. PROVIDE CHECK VALVE AND AUTOMATIC BALL DRIP.
4. PROVIDE AUXILIARY DRAINS PER NFPA 13 8.16.2 TO FACILITATE DRAINAGE OF TRAPPED PIPING.
5. 2" DROP TO SUPPLY PIPING BELOW MEZZANINE. INSTALL TIGHT TO WALL. CONNECT TO CROSS MAIN BELOW ROOF DECK.
6. HYDRAULIC DESIGN AREA #1 - VEHICLE MAINTENANCE AREA, ORDINARY HAZARD GROUP II - 0.20 GPM/SQ FT OVER 1,950 SQ FT (AREA INCREASED 30% FOR ROOF SLOPE) WITH 250 GPM HOSE ALLOWANCE. SYSTEM REQUIRES 23.7 PSI WITH 469 GPM FLOWING AT BASE OF FIRE SPRINKLER RISER.
7. INSTALL SWING JOINT ON PIPING NEAR ROOF PEAK TO ALLOW PIPING TO BE RUN PARALLEL TO ROOF SLOPE
8. INSTALL ADDITIONAL FIRE SPRINKLERS TO PROTECT BELOW OVERHEAD DOORS (NOT SHOWN ON PLAN) IN ACCORDANCE WITH NFPA 13 8.5.5.3.
9. INSTALL DRY SIDEWALL FIRE SPRINKLER TO PROTECT BELOW EXTERIOR CANOPY WITH THE POTENTIAL FOR COMBUSTIBLE STORAGE BELOW. INSTALL SPRINKLER WITH DEFLECTOR APPROXIMATELY 6" BELOW AND PARALLEL TO ROOF DECK.
10. HYDRAULIC DESIGN AREA #2 - OFFICE SPACE, LIGHT HAZARD - 0.10 GPM/SQ FT OVER 950 SQ FT (AREA REDUCED PER USE OF QR SPRINKLERS) WITH 100 GPM HOSE ALLOWANCE. SYSTEM REQUIRES 34.8 PSI WITH 197.2 GPM FLOWING AT BASE OF FIRE SPRINKLER RISER.
11. WATER FLOW BELL. INSTALL ABOVE FIRE DEPARTMENT CONNECTION. PROVIDE 120VAC POWER SUPPLY TO BELL SWITCHED THROUGH WATER FLOW SWITCH AT FIRE SPRINKLER RISER.

FIRE SPRINKLER SYMBOL LEGEND

SYMBOL	DESCRIPTION
○	QUICK RESPONSE, STANDARD ORIFICE (K-5.6), ORDINARY TEMPERATURE, BRASS, UPRIGHT FIRE SPRINKLER ON EXPOSED BRANCH LINE
⊘	QUICK RESPONSE, STANDARD ORIFICE (K-5.6), ORDINARY TEMPERATURE, CHROME PLATED, PENDENT FIRE SPRINKLER WITH CHROME, RECESSED TYPE ESCUTCHEON
⊘	STANDARD RESPONSE, LARGE ORIFICE (K-8.0), INTERMEDIATE TEMPERATURE, BRASS, UPRIGHT FIRE SPRINKLER ON EXPOSED BRANCH LINE
◁	QUICK RESPONSE, STANDARD ORIFICE (K-5.6), ORDINARY TEMPERATURE, CHROME PLATED, SIDEWALL FIRE SPRINKLER ON EXPOSED BRANCH LINE
◀	QUICK RESPONSE, STANDARD ORIFICE (K-5.6), ORDINARY TEMPERATURE, CHROME PLATED, DRY SIDEWALL FIRE SPRINKLER WITH CHROME ESCUTCHEON
◯	WET-PIPE FIRE SPRINKLER RISER
	WALL MOUNT FIRE DEPARTMENT CONNECTION
	AUXILIARY DRAIN
—○—	VERTICAL TRANSITION FOR PIPING
—○—	VERTICAL RISE FROM HORIZONTAL PIPE
—●—	VERTICAL DROP FROM HORIZONTAL PIPE
—▮—	PIPE CAP/FLUSHING CONNECITON
ⓧ	HYDRAULIC CALCULAITON NODE POINT
X'-X" AFF/BD	PIPING ELEVATION ABOVE FINISHED FLOOR OR BELOW DECK

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06/05/23

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PROJECT	DATE
22123.00	06/05/23
DRAWN	CHECKED
NMJ	GTJ

REVISED

SHEET TITLE
FIRE SPRINKLER PLAN - MAIN LEVEL

SHEET
FS-01

ORIGINAL SHEET SIZE
30" x 42"